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ChronoChart

Setup-Agnostic Channel Charting for Label-Free Indoor Localization

Majid Sharghi Foroushani, Jonas Pirkel, Jonathan Ott, Maximilian Stahlke, George Yammine, Tobias Feigl
and Christopher Mutschler

Motivation

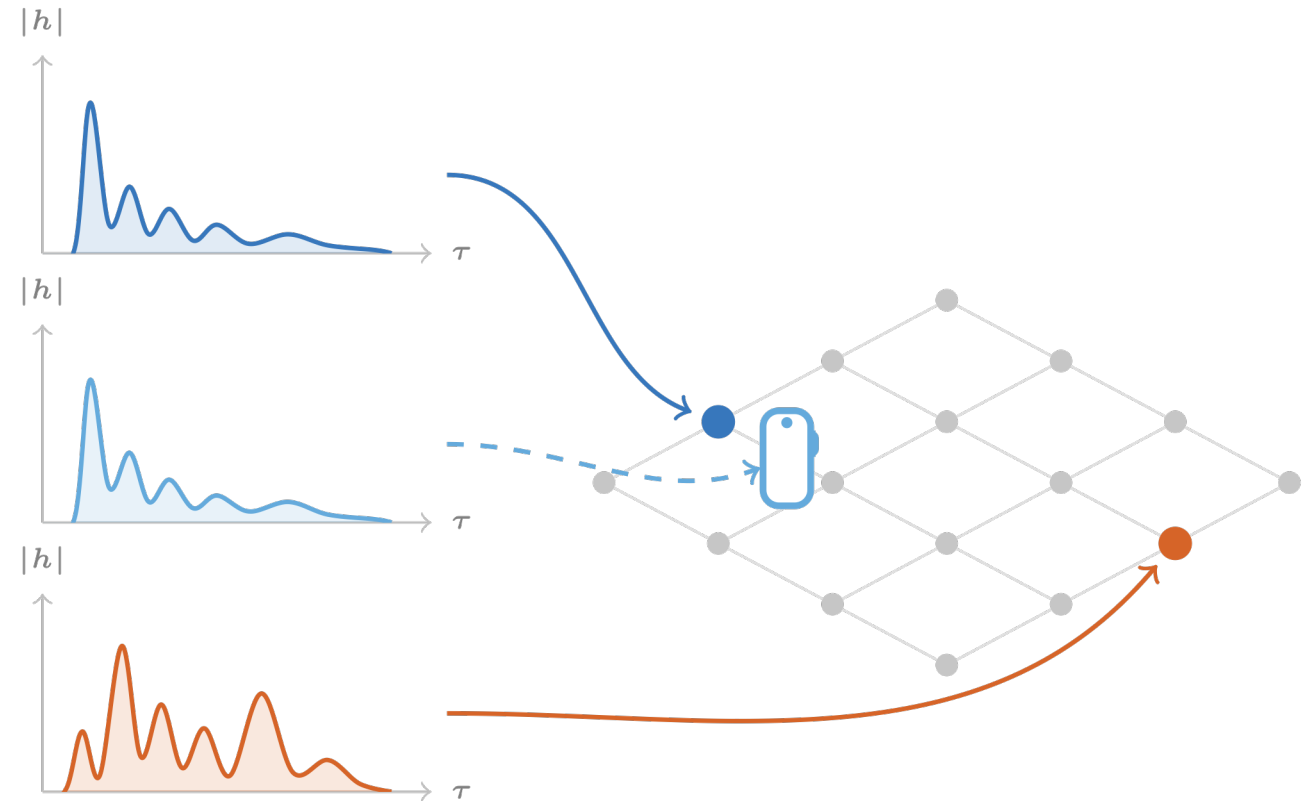
Why Channel Charting

Classic radio-based localization

- Challenging under NLOS and multipath conditions

Supervised fingerprinting

- High accuracy
- Requires labeled data
- Retraining on environment changes



Motivation

Why Channel Charting

Classic radio-based localization

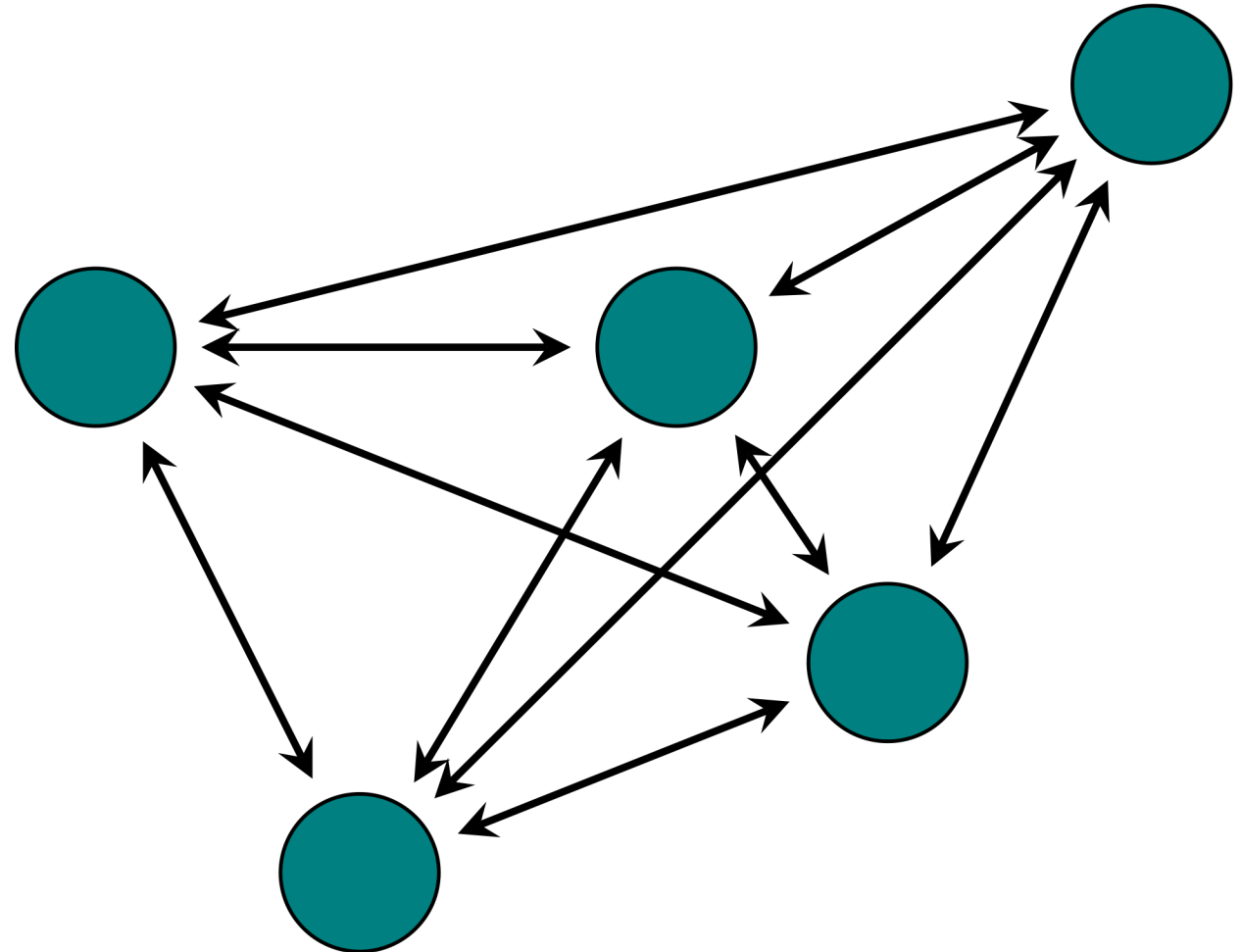
- Challenging under NLOS and multipath conditions

Supervised fingerprinting

- High accuracy
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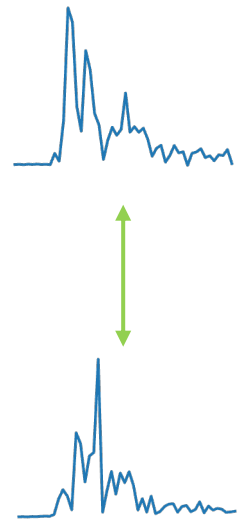
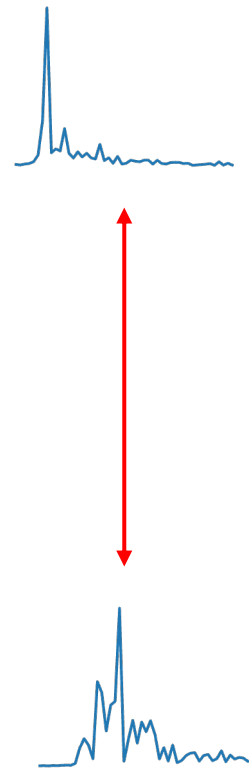
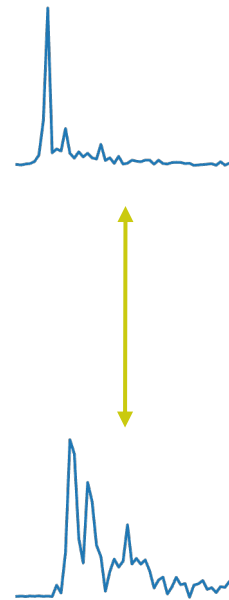
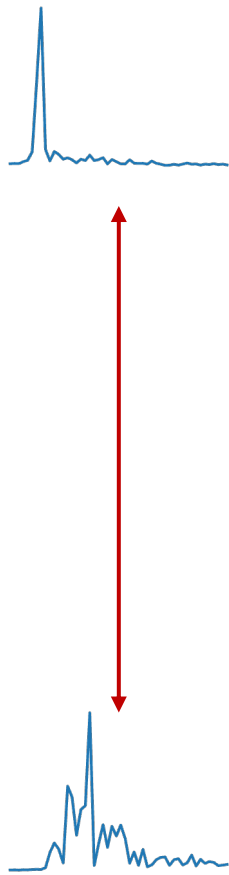
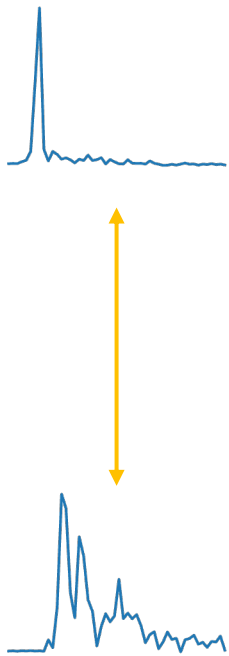
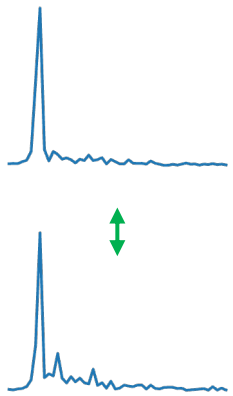
Channel Charting

- Mapping channel measurements $\rightarrow$ pseudo-positions
- Self-Supervised



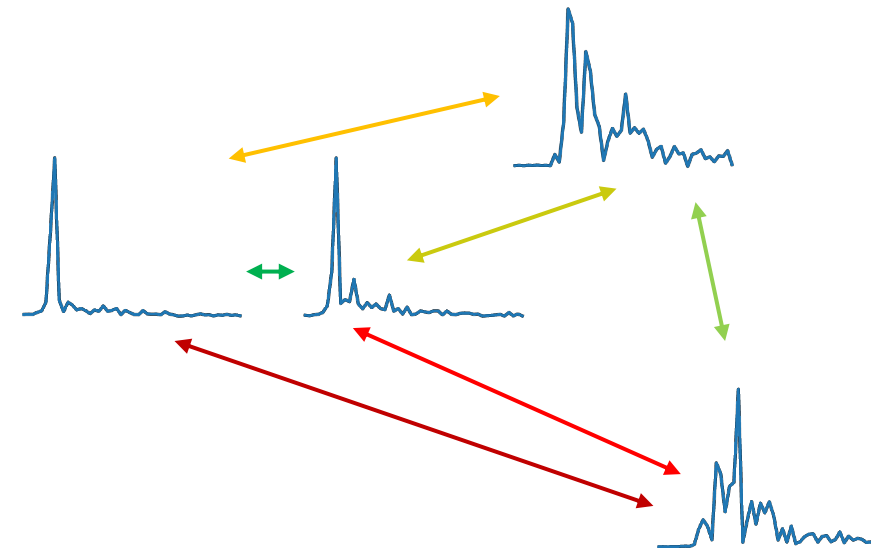
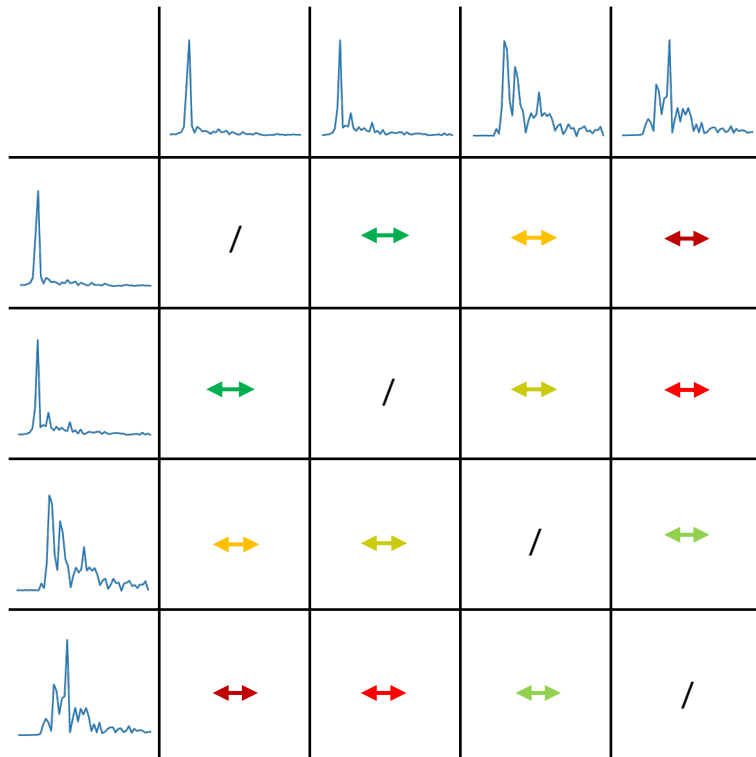
Motivation

Why Channel Charting



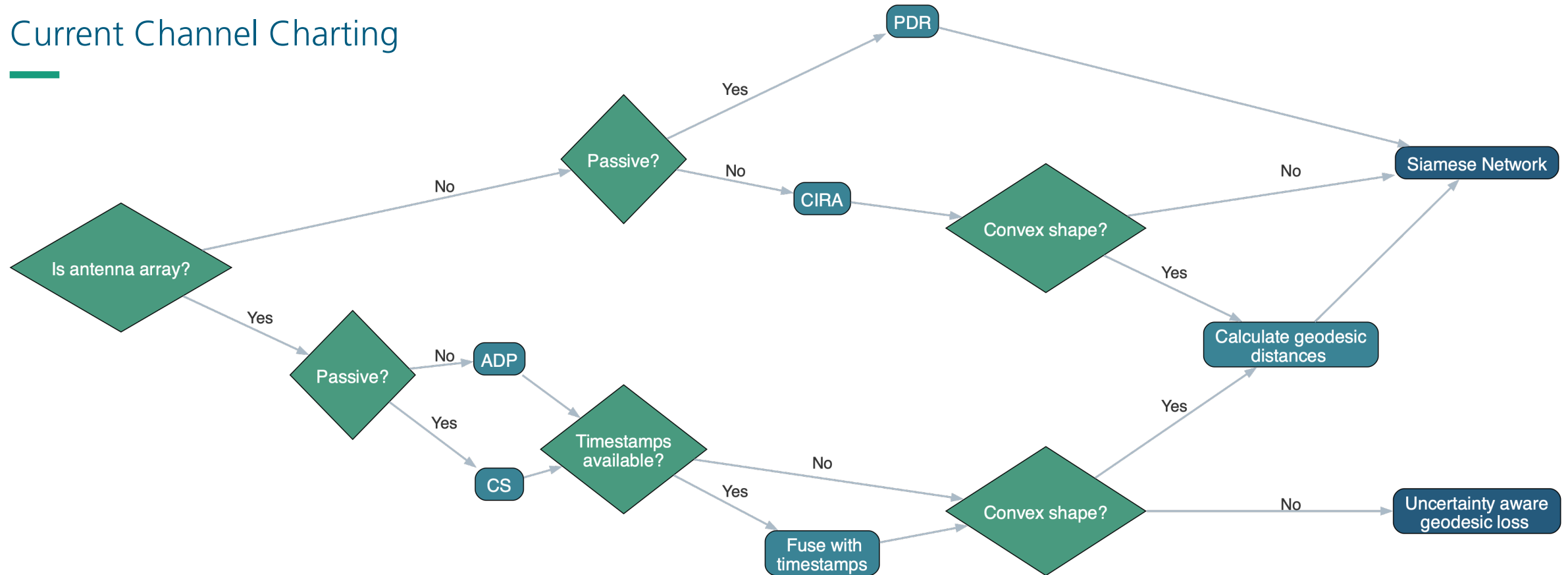
Motivation

Why Channel Charting



Motivation

Current Channel Charting



n^2 Memory requirement

- 10 min ~ 14 GB*
- 30 min ~ 130 GB*
- 1h ~ 520 GB*

*100 Hz recording rate, fp32 for distances

ChronoChart

No setup specific information, just timestamps, with linear complexity

Loss functions

Three objectives protect the latent chart

GLOBAL STRUCTURE

Triplet loss

Anchor-positive pairs are pulled together while temporally distant negatives are pushed apart.

 $\mathcal{L}_{\text{triplet}}$

TEMPORAL SMOOTHNESS

Temporal-proximity loss

Embedding distances follow timestamp differences, suppressing unnecessary jumps in a trajectory.

 $\mathcal{L}_{\text{time}}$

DATA-DRIVEN PROXIMITY

Joint-embedding loss

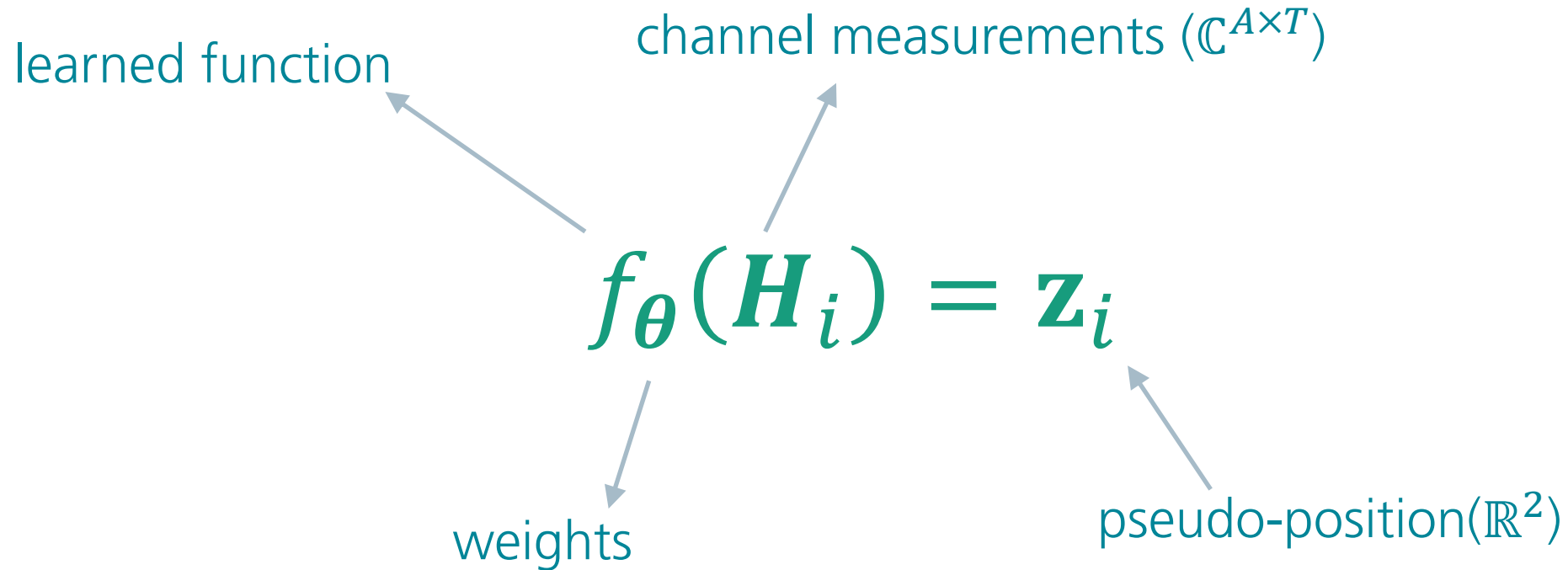
Original and masked views must agree in latent space

 $\mathcal{L}_{\text{JE}}$

NashMTL balances all three gradients at every update.

ChronoChart

Temporal Supervision



ChronoChart

Temporal Supervision

Δz : distance in position

Δt : distance in time

$$\mathcal{L}_{time} = w(|\mathbf{z}_i - \mathbf{z}_{i+1}| - k|t_i - t_{i+1}|)^2$$

samples closer in time
get more weight

$$w = 1 - \frac{2\Delta t}{T}$$

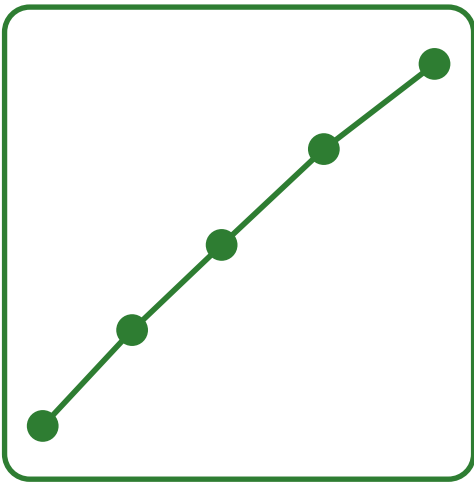
$$\forall \Delta t < T$$

learnable factor

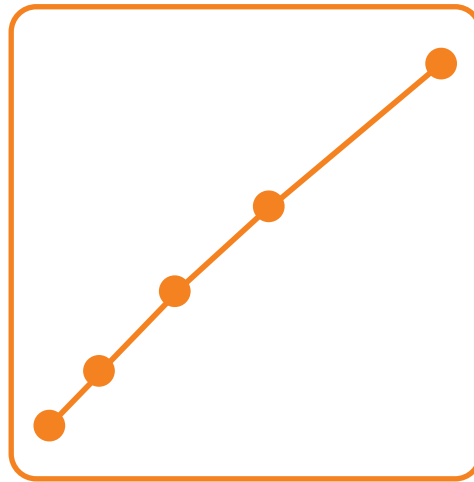
time threshold

ChronoChart

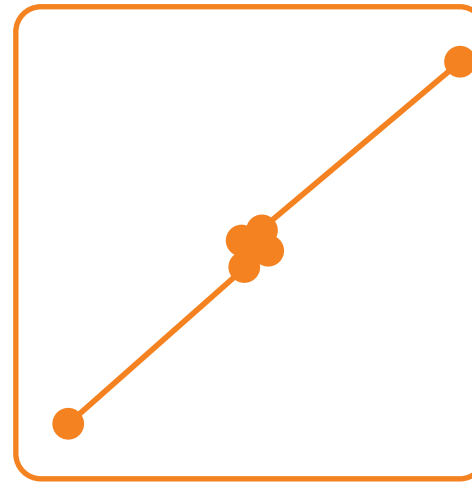
Temporal Supervision - Problems



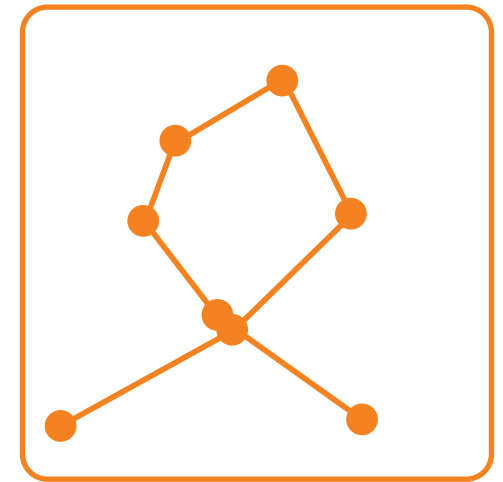
Continuous motion



Variable speed



Stationary periods



Trajectory crossings

$$\mathcal{L}_{\text{JE}} = \|\mathbf{z}_i - \hat{\mathbf{z}}_i\|_2^2$$

$\mathbf{z}_i = f_{\theta}(\mathbf{H}_i)$

$\hat{\mathbf{z}}_i = f_{\theta}(\text{augment}(\mathbf{H}_i))$

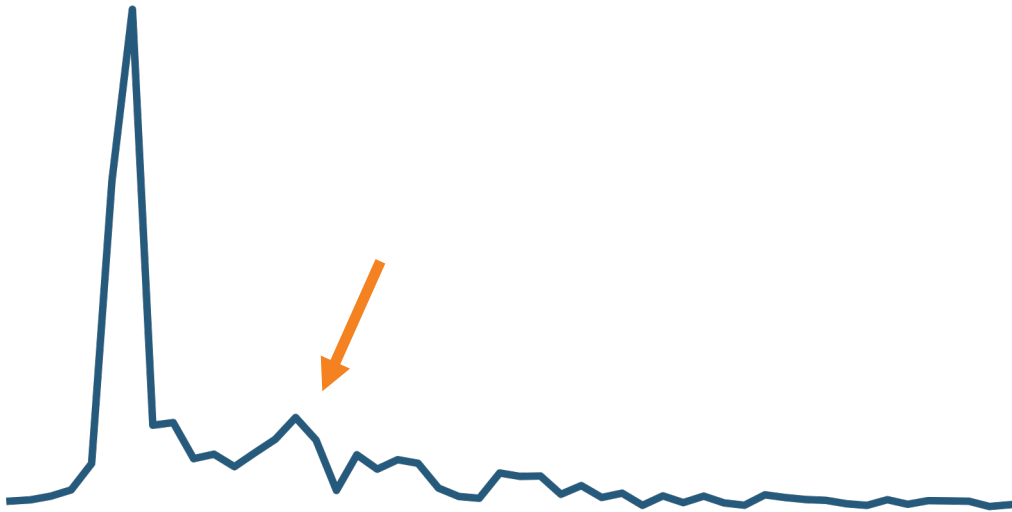
Method

Channel Charting



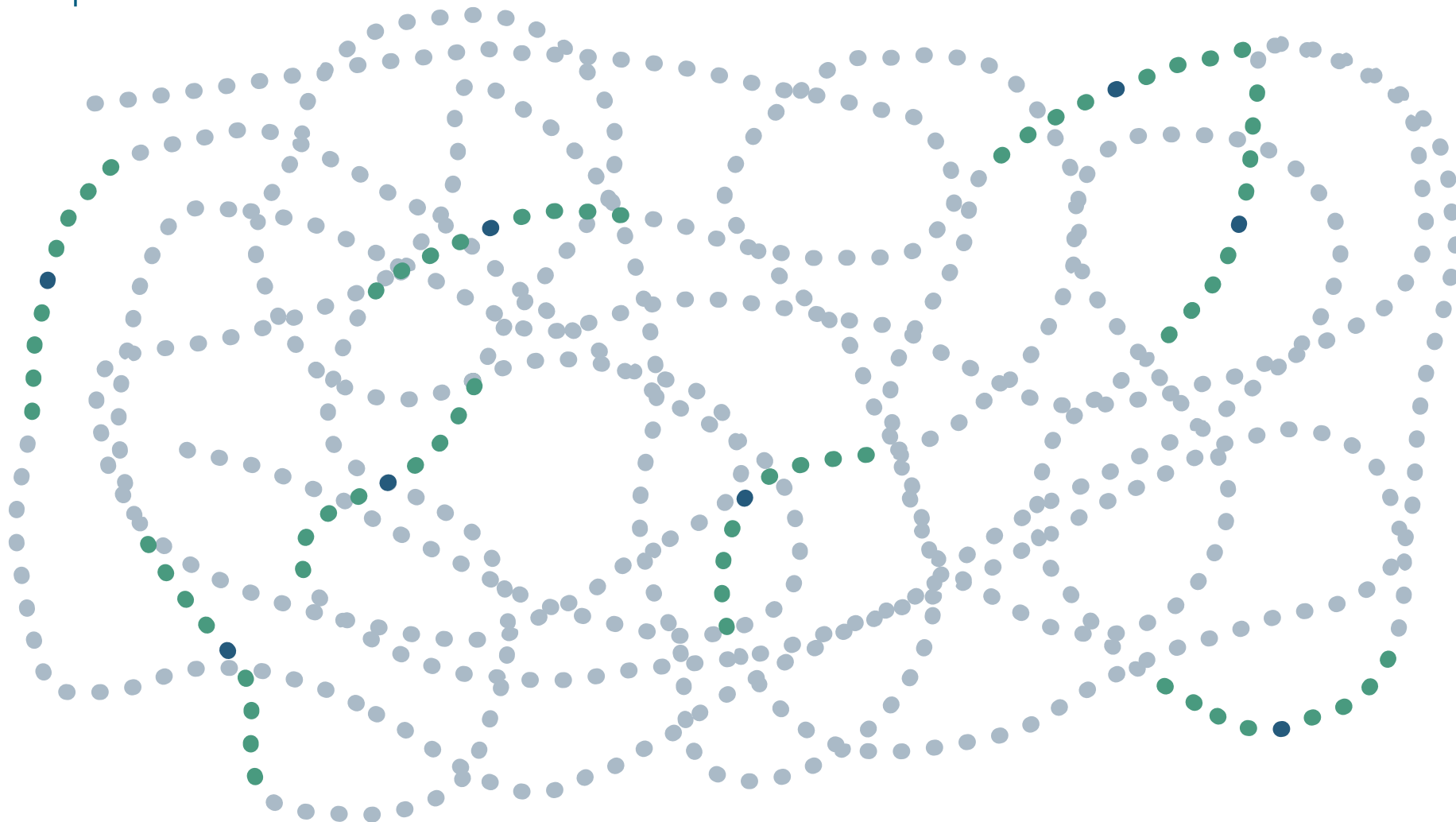
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Channel Charting



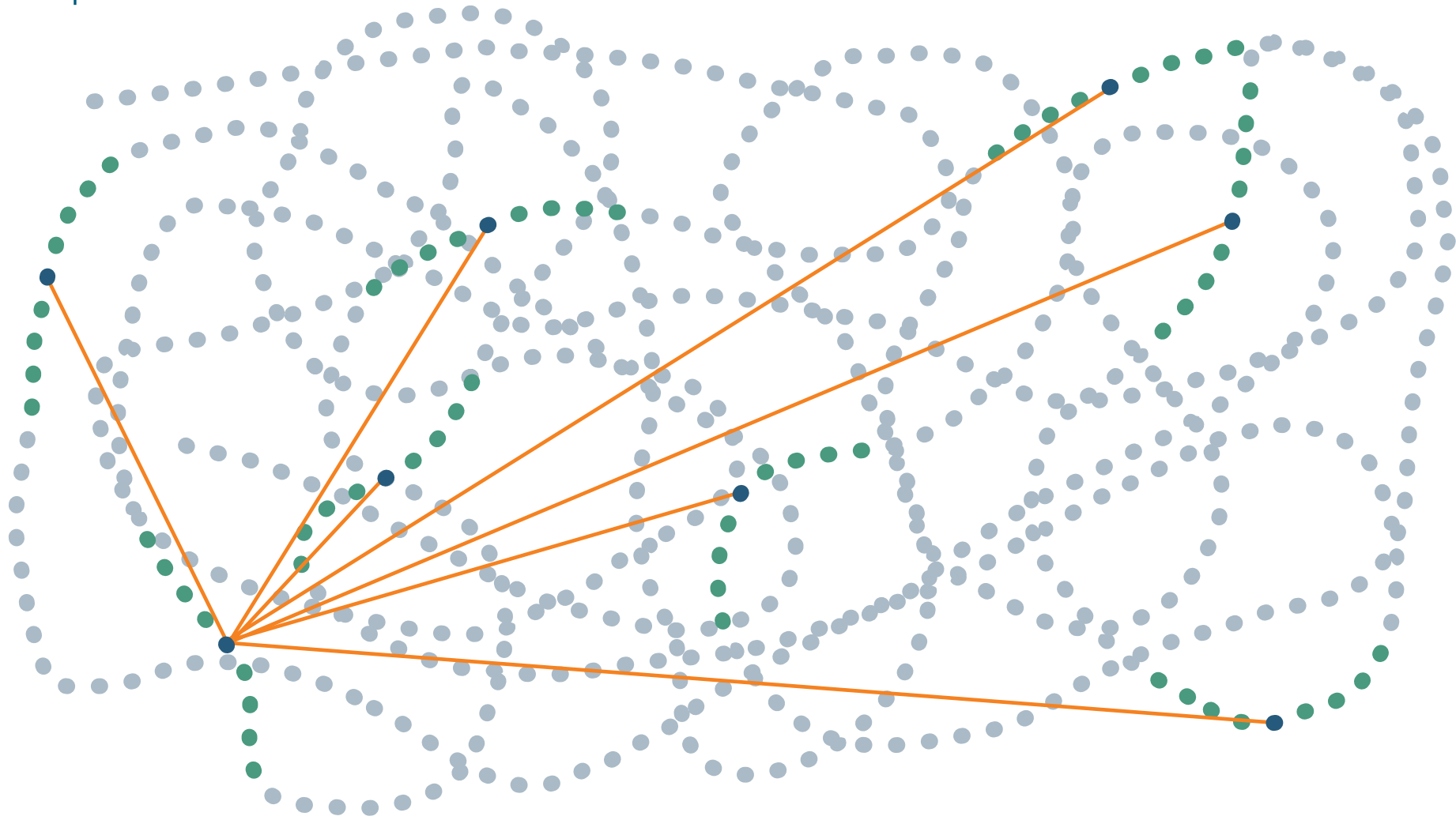
ChronoChart

Global relationships



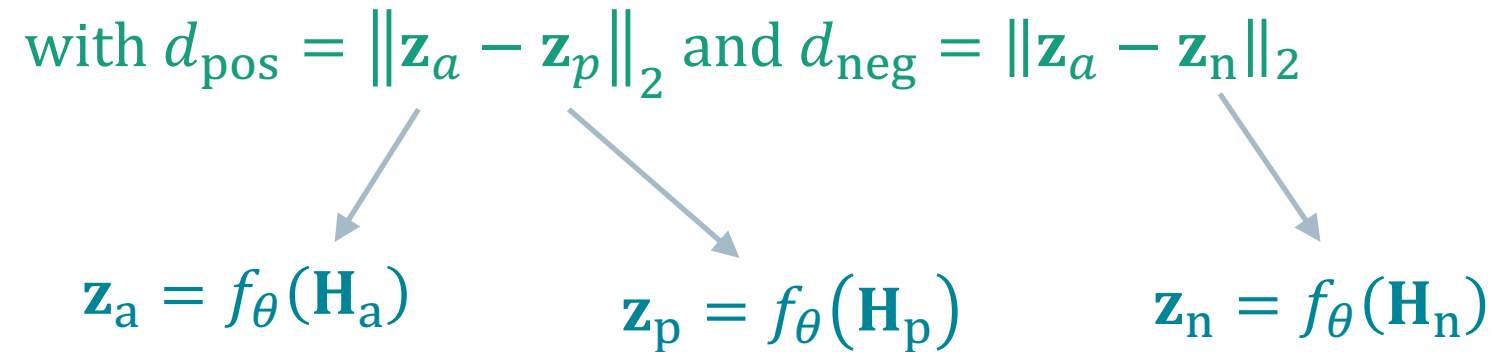
ChronoChart

Global relationships



$$\mathcal{L}_{\text{triplet}} = -\log \left(\frac{\exp(-d_{\text{pos}})}{\exp(-d_{\text{pos}}) + \exp(-d_{\text{neg}})} \right)$$

with $d_{\text{pos}} = \|\mathbf{z}_a - \mathbf{z}_p\|_2$ and $d_{\text{neg}} = \|\mathbf{z}_a - \mathbf{z}_n\|_2$

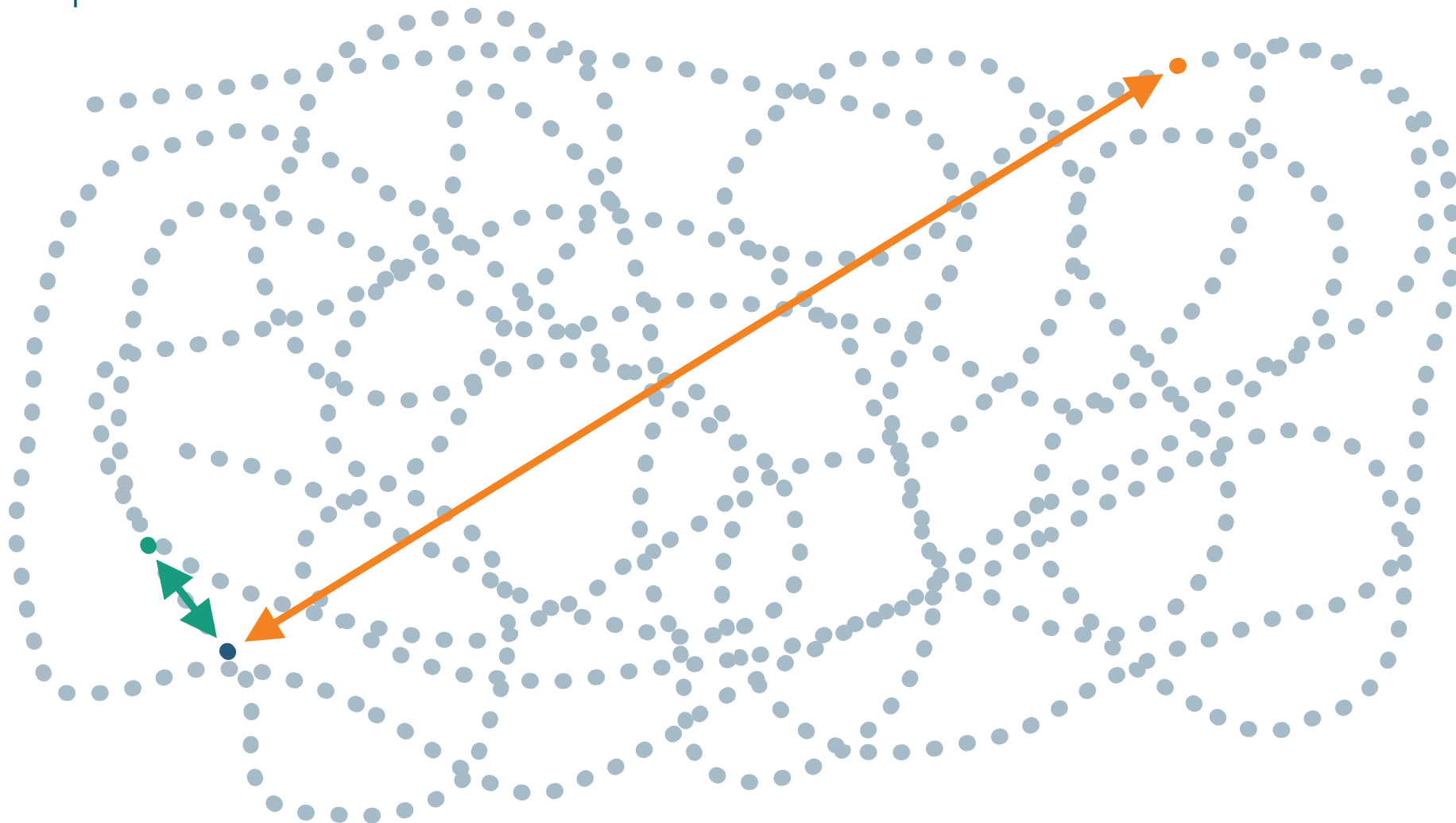

$$\mathbf{z}_a = f_{\theta}(\mathbf{H}_a)$$

$$\mathbf{z}_p = f_{\theta}(\mathbf{H}_p)$$

$$\mathbf{z}_n = f_{\theta}(\mathbf{H}_n)$$

ChronoChart

Global relationships



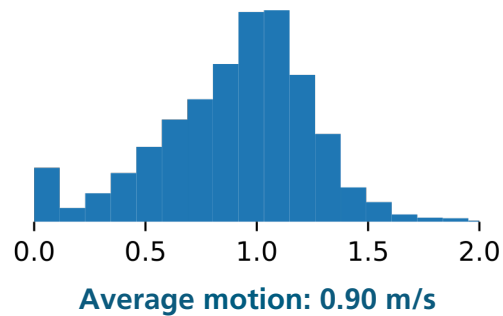
Datasets

Diverse setups, same method

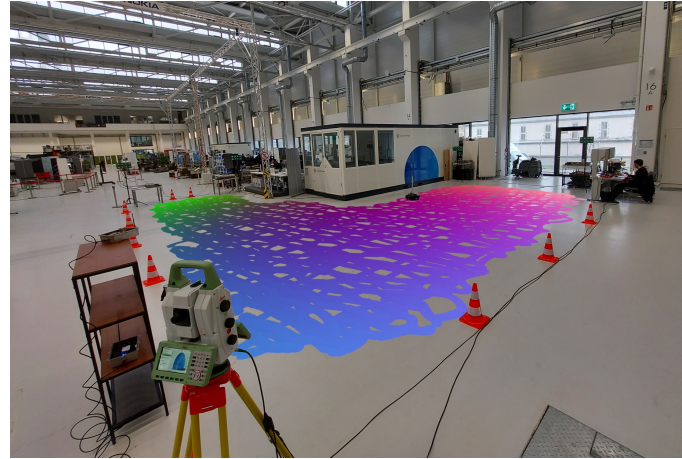
Fraunhofer 5G



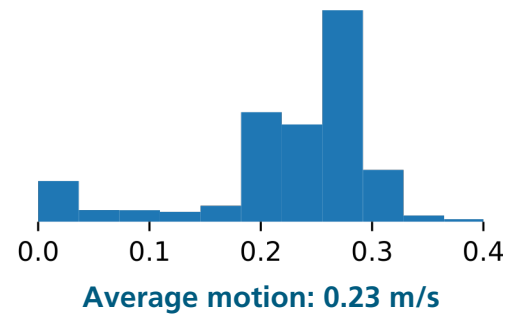
Distributed single antenna 5G



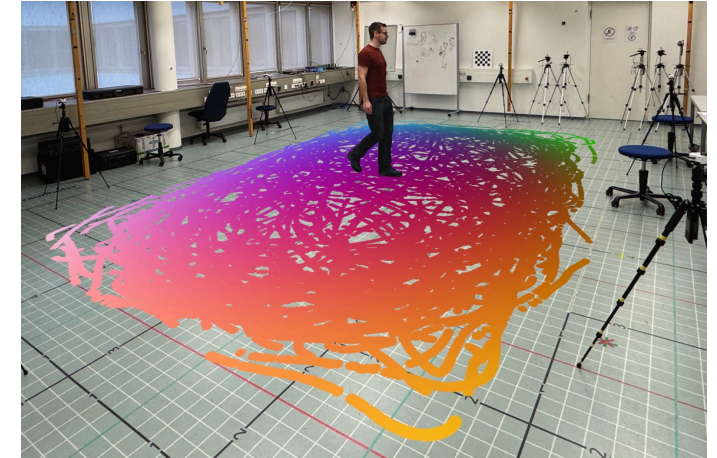
Dichasus



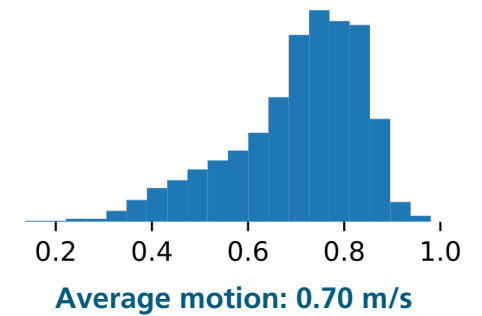
Distributed massive MIMO



Passive UWB



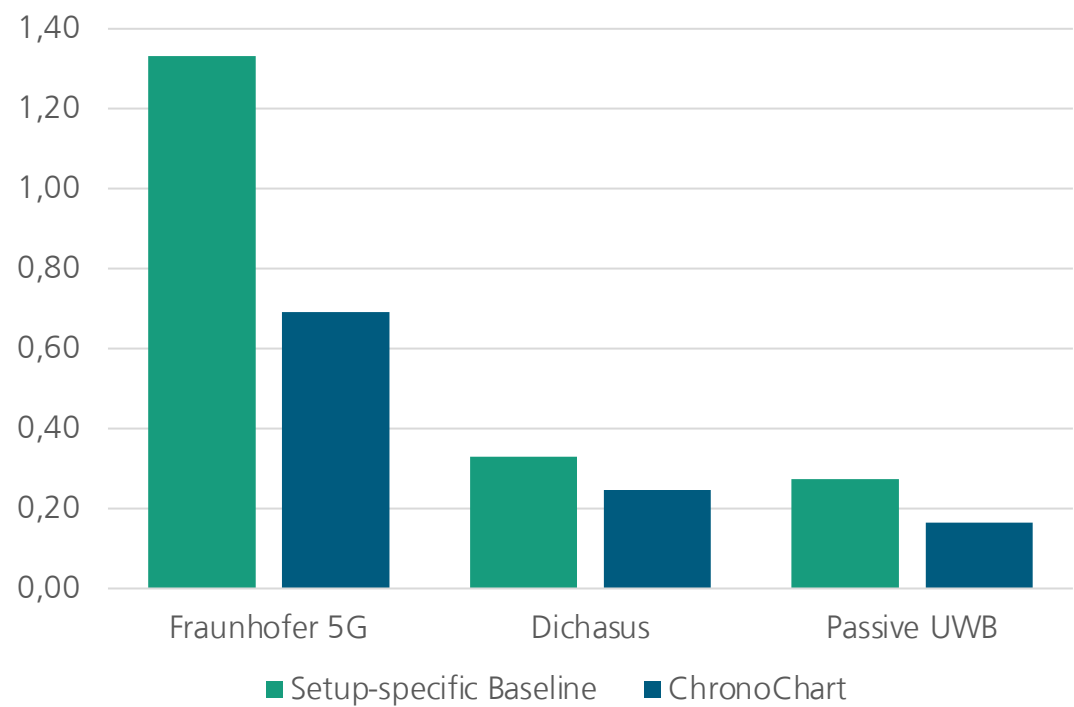
Passive single antenna UWB



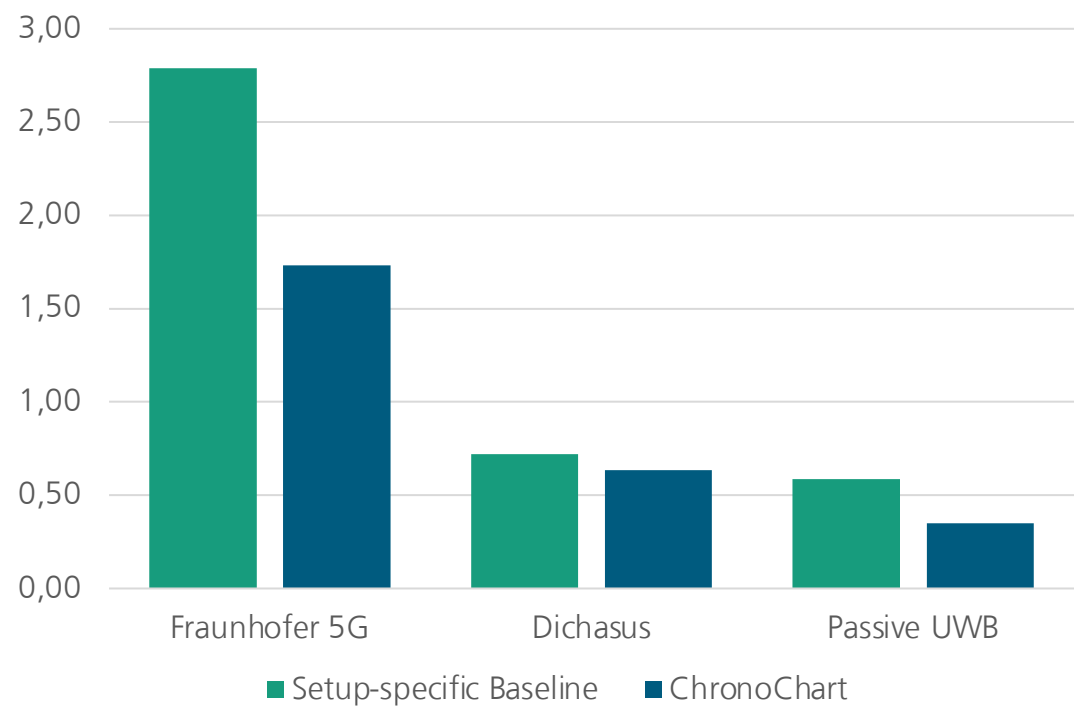
Results



Mean Average Error

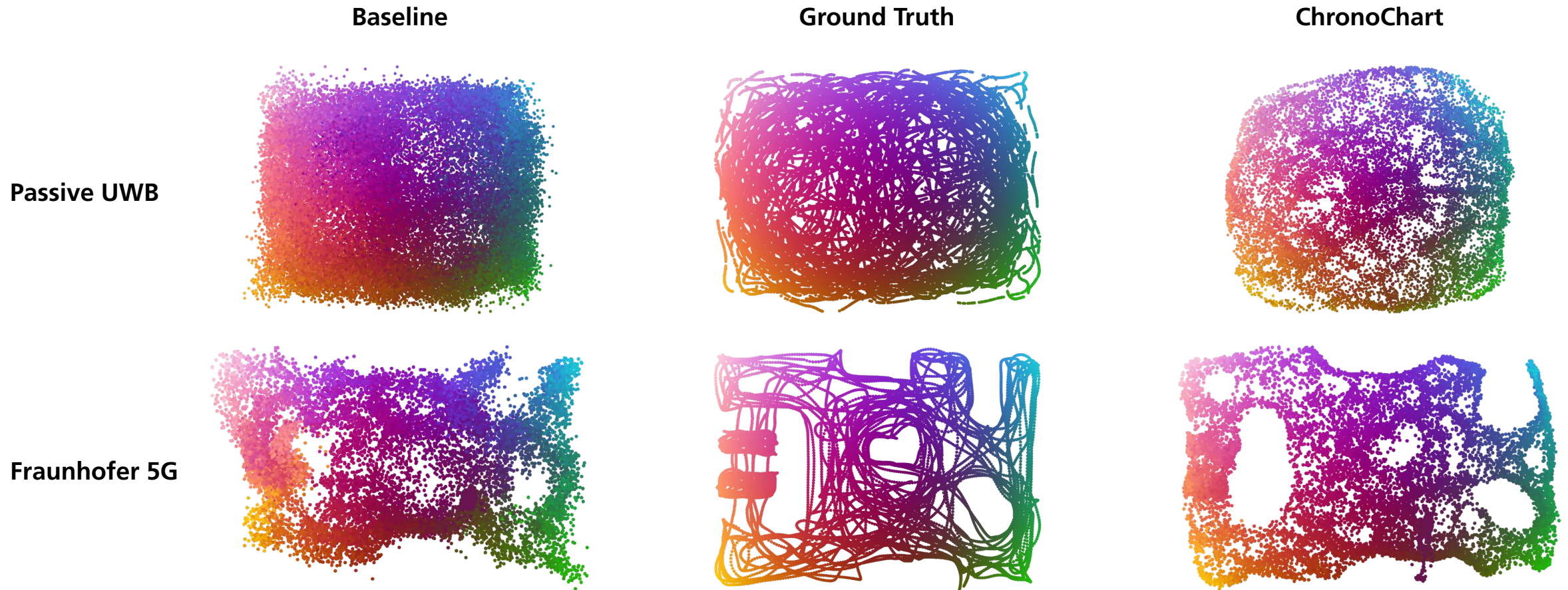


95th percentile



Results

Channel Charts



Results

Channel Charts - Dichasus

Baseline



Ground Truth



ChronoChart



Conclusion

ChronoChart

ChronoChart

- Only needs timestamps
- Setup agnostic
- No distance matrices
- Still better than baselines

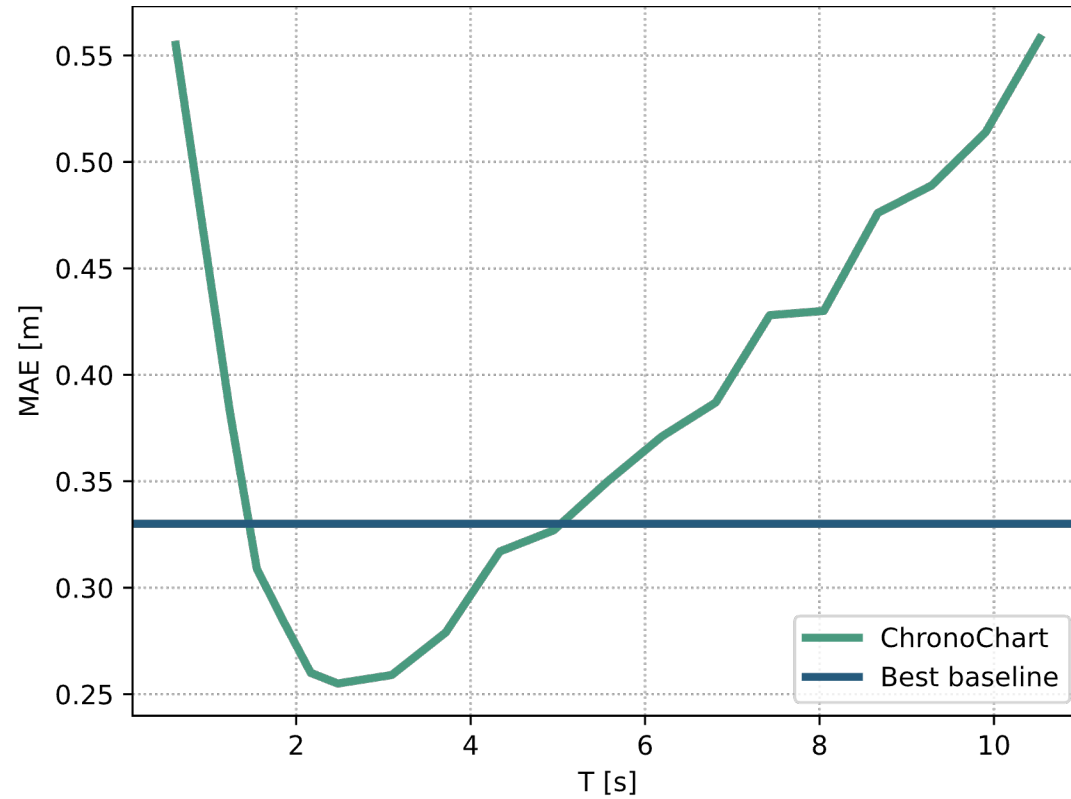
Code Available

- Try out on your own dataset!

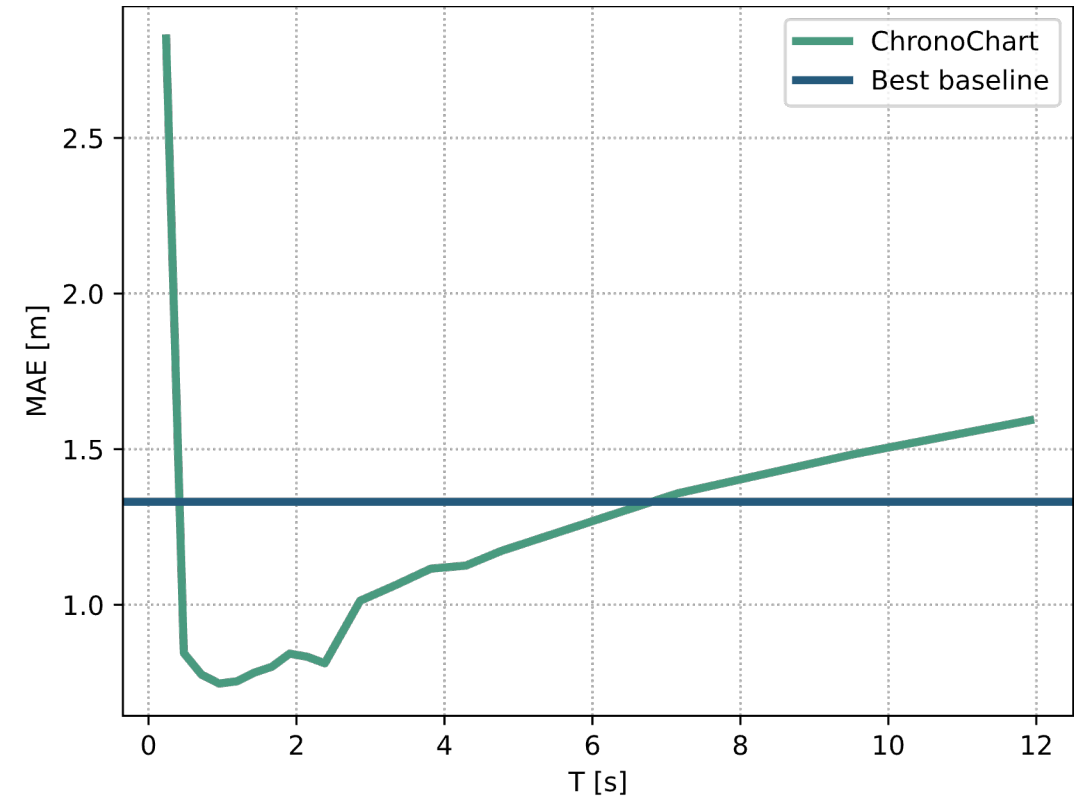
Ablation Study

Time Threshold T

Dichasus



Fraunhofer 5G



Ablation Study

Loss Coefficients

Fraunhofer 5G — error (m)

		Temporal coefficient λ_2			
		0.0	0.5	1.0	1.5
Joint Embedding coefficient λ_3	0.0	2.18	0.99	1.04	0.97
	0.5	1.01	0.94	0.98	0.98
	1.0	1.01	0.84	0.98	0.94
	1.5	1.14	0.86	0.98	0.86

Dichasus — error (m)

		Temporal coefficient λ_2			
		0.0	0.5	1.0	1.5
Joint Embedding coefficient λ_3	0.0	1.23	0.62	0.80	0.55
	0.5	0.66	0.40	0.48	0.40
	1.0	0.48	0.38	0.38	0.39
	1.5	0.66	0.41	0.39	0.39

green = lower error red = higher error